The Harvard Computers

Women who measured the stars — before computer meant a machine

Abbé Faria lesson for Jeff Franzen · 6 November 2026 · faria.jbf.com/topics/51-harvard-computers/

Jeff, reading about Cepheids on the Polaris page, stopped at "Henrietta Swan Leavitt (1868–1921), a Harvard 'computer'" and said he could not be left hanging on the word. This lesson is the full answer.



Women astronomical computers at Harvard College Observatory, c. 1891 — "Pickering's Harem." Williamina Fleming near center; Antonia Maury (with glass) third from left. Leavitt and Cannon are not in this photo. Harvard College Observatory / Public domain.

1. The word: computer meant a person

Computer is compute + -er: a person who computes.

Earliest recorded English use: Richard Brathwaite (also spelled Braithwaite), The Yong Mans Gleanings (London, 1613):[1]

> *"I haue read the truest computer of Times, and the best Arithmetician that euer breathed, and he reduceth thy dayes into a short number."*

The OED's historical sense: a person employed to make calculations in an observatory, in surveying, etc. The electronic-machine sense clusters in the 1940s. Until then, computer almost always meant a human being.

Gloss. In Brathwaite the "computer" is God, the reckoner of our days ("The daies of Man are threescore and ten"); the word then moved from a reckoner of time, to human calculators like the Harvard women, to machines.

[1] Footnote — Brathwaite 1613. Exact wording as cited by the Oxford English Dictionary (earliest quotation for computer): Richard Brathwaite ("R. B."), The Yong Mans Gleanings (London, 1613), p. 1: "I haue read the truest computer of Times, and the best Arithmetician that euer breathed, and he reduceth thy dayes into a short number." Original spelling preserved (haue, euer, dayes). The passage continues: "The daies of Man are threescore and ten" (Psalm 90:10) — so the "computer of Times" is God counting out a human life, not a clerk with an abacus. Same quotation in Wikipedia's Human computer etymology note.

Human computers before Harvard. The British Nautical Almanac hired dozens of human computers for lunar and solar tables. From 1765–1811 about 35 worked on it; Mary Edwards was a woman who computed for a living. That world is the cousin of Nathaniel Bowditch's world of checked tables a navigator trusted with a ship.

2. Edward Pickering and the factory observatory

Edward Charles Pickering (1846–1919) became director of the Harvard College Observatory in 1877. Photography and spectroscopy flooded astronomy with glass plates. Starting in the early 1880s (Fleming moved from household staff into the computing room in 1881), he hired women to measure positions, magnitudes, and spectra by day. Over the decades about 80 women served. They were called computers. Observing itself stayed mostly closed to them.

Pay was about 25 cents an hour (Fleming as curator ~50¢; Leavitt reportedly ~30¢). Men doing similar work earned substantially more. The mocking nickname was "Pickering's Harem." Modern writers say the Harvard Computers.

The Observatory (Cambridge, plus Arequipa, Peru) amassed on the order of 500,000 glass plates. Funding came largely from Anna Palmer Draper, widow of Henry Draper — the Henry Draper Memorial that paid for the spectral survey and the Henry Draper Catalogue (HD).



Computers at work over astronomical photographs, c. 1890. Public domain.

3. Williamina Fleming: from housemaid to curator



Williamina Paton Stevens Fleming, c. 1890s. Public domain.

Williamina Paton Stevens Fleming (1857–1911), a Scottish schoolteacher, emigrated to Boston, was abandoned while pregnant, and worked as Pickering's maid/housekeeper (~1879). In 1881 he put her on the Observatory payroll. She helped invent the early Harvard spectral letters; discovered the Horsehead Nebula on a plate taken 1888 Feb 6 by W. H. Pickering (she noted it 1888 May 19); found hundreds of variables, novae, and the first recognized white dwarfs; and in 1899 became Harvard's first Curator of Astronomical Photographs.

4. Antonia Maury: a finer classification

Antonia Maury (1866–1952), a niece of Henry Draper, built a finer scheme that distinguished giants from dwarfs by line width. Pickering preferred Fleming's simpler letters for the big catalogue; Maury's subtler distinctions were vindicated when Hertzsprung and Russell made luminosity classes central to stellar evolution.

Oh Be A Fine Girl, Kiss Me

How Annie Jump Cannon sorted the stars.

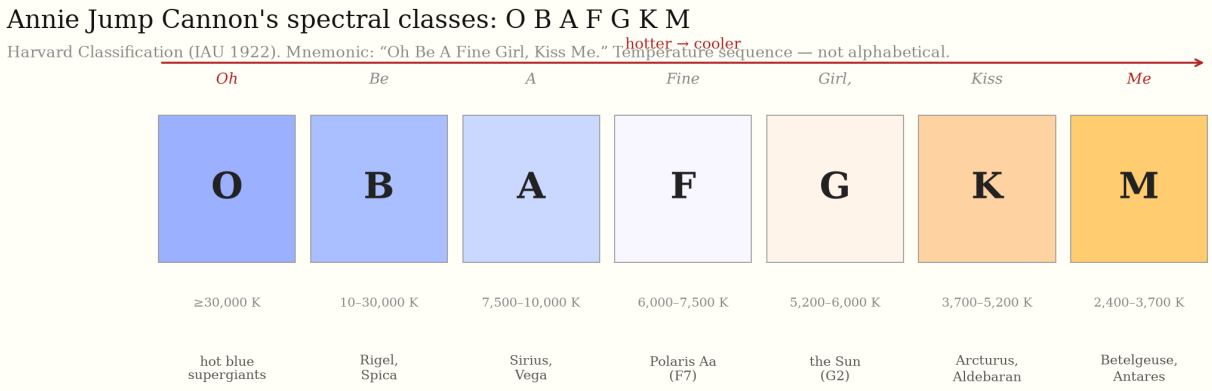


Annie Jump Cannon, 1922. Public domain.

Annie Jump Cannon (1863–1941) studied at Wellesley, lost much of her hearing to scarlet fever, and joined in 1896. She reordered Fleming's letters into the temperature sequence still used worldwide:

O – B – A – F – G – K – M

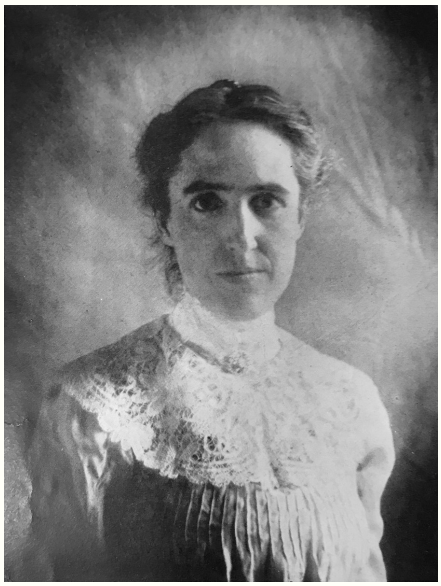
That mnemonic — the heading of this section — looks alphabetical; it is a temperature sequence (O hottest, M coolest). The Sun is G2. Polaris Aa is F7Ib.



Cannon's spectral sequence with approximate temperatures and familiar examples. Diagram for this lesson.

For the nine volumes of the Henry Draper Catalogue (1918–24) she classified about 225,300 spectra; across her lifetime about 350,000 (some tallies near 400,000). The IAU adopted her system in 1922. She received the Henry Draper Medal (1931) but was not made a Harvard professor until 1938.

6. Henrietta Swan Leavitt: the period–luminosity law



Henrietta Swan Leavitt (1868–1921). Public domain.

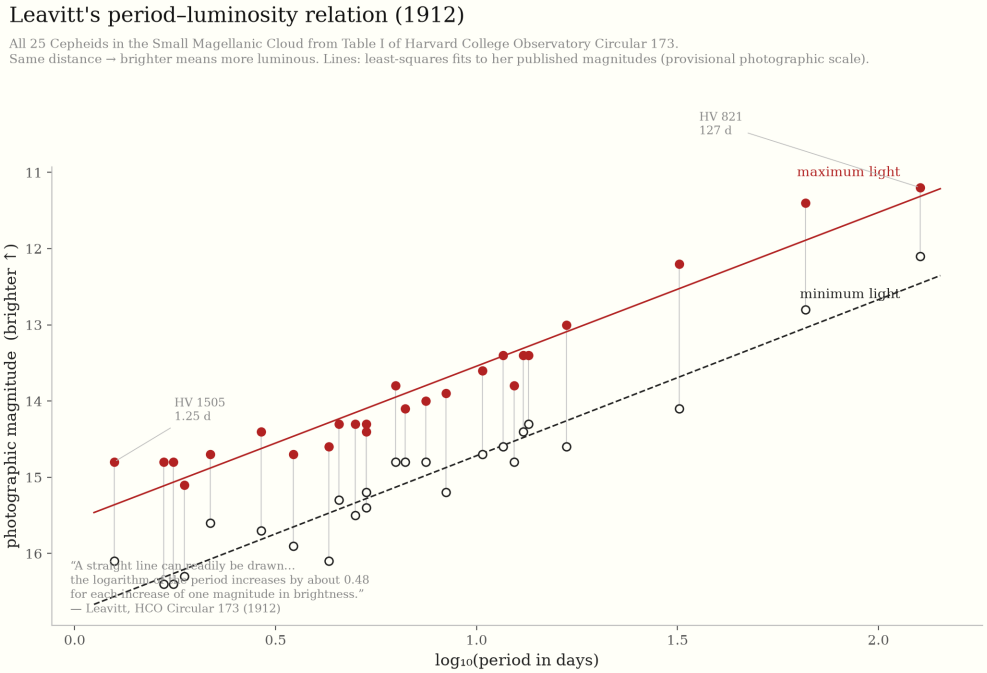
Henrietta Swan Leavitt (4 Jul 1868 – 12 Dec 1921) graduated from what became Radcliffe, volunteered in 1895, took a permanent post in the early 1900s, and was nearly deaf. Her assignment: variable stars on Magellanic Cloud plates.

1908. In 1777 Variables in the Magellanic Clouds she wrote:

> *"It is worthy of notice that the brighter variables have the longer periods."*

1912. HCO Circular 173 (3 Mar 1912), prepared by Miss Leavitt under Pickering's signature, plotted 25 Cepheids:

> *"A straight line can readily be drawn among each of the two series of points corresponding to maxima and minima… The logarithm of the period increases by about 0.48 for each increase of one magnitude in brightness."*



All 25 stars from Table I of Circular 173 (1912), replotted from her published periods and provisional magnitudes. Brighter is upward.

If period tells true brightness, comparing true with apparent brightness yields distance. Cepheids become standard candles. Hertzsprung (1913) and Shapley calibrated the zero point; Edwin Hubble (1923–25) used Cepheids in

Andromeda to show it lies outside the Milky Way. The relation is now often called Leavitt's Law.

The Nobel letter that came too late. On 23 Feb 1925, Gösta Mittag-Leffler wrote, unaware she had died in 1921:

> *"Honoured Miss Leavitt… I feel seriously inclined to nominate you to the Nobel prize in physics for 1926…"*

Nobels are not awarded posthumously. She never received the nomination.



Annie Jump Cannon (left) and Henrietta Swan Leavitt (right), 1913. Public domain.

7. Cecilia Payne: stars are mostly hydrogen

Cecilia Payne (later Payne-Gaposchkin, 1900–1979) came from Cambridge to Harvard in 1923. Her 1925 Ph.D. thesis used Cannon's classes and Saha ionization theory to show stars are overwhelmingly hydrogen and helium. A senior reviewer urged her to soft-pedal; she did in print, then lived to see it become orthodoxy. In 1956 she became Harvard's first woman professor of astronomy.

8. Later recognition

- Dava Sobel, The Glass Universe (2016).
- DASCH and Project PHaEDRA: digitizing ~500,000 plates and the computers' notebooks.
- Lunar crater Leavitt; others honor Cannon and Fleming. AAS (2008) urged calling the relation Leavitt's Law.

Review questions

1. What did computer mean in 1613, and who used it?
2. Rough pay per hour, and the mocking nickname?
3. Who found the Horsehead Nebula, on whose plate, what year?
4. Recite O B A F G K M and the mnemonic. Sun's letter? Polaris Aa?
5. What did Leavitt notice in 1908, and what did Circular 173 prove?

6. Why does a shared distance make a period–luminosity relation possible?
7. What did Mittag-Leffler write in 1925, and why no Nobel?
8. What did Payne's 1925 thesis conclude?

Answers

1. A person who calculates; Richard Brathwaite, Yong Mans Gleanings (1613), "the truest computer of Times" — first of God, later of human calculators, then of machines.
2. ~25¢/hour; "Pickering's Harem."
3. Williamina Fleming, on W. H. Pickering's plate, 1888.
4. O B A F G K M — "Oh Be A Fine Girl, Kiss Me." Sun = G (G2). Polaris Aa = F (F7Ib).
5. Brighter Magellanic variables have longer periods; 1912: straight-line log(period)–brightness relation for 25 Cepheids.
6. Same distance → apparent brightness differences are true luminosity differences.
7. He was inclined to nominate her for the 1926 Physics Nobel; she had died in 1921.
8. Stars are mostly hydrogen and helium.

Companion page: https://faria.jbf.com/topics/51-harvard-computers/ · Related: Polaris (48), Bowditch (49), Sidereal day (50).